

RESEARCH REPORT — DECISION INTELLIGENCE

Factory Reallocation & Shipping Optimization Recommendation System

Nassau Candy Distributor — Predictive Logistics & Factory Assignment Intelligence

Organization: **Nassau Candy Distributor**
Submitted to: **Unified Mentor**
Dataset: 10,194 orders | 15 products | 5 factories
Year: 2024

1. Executive Summary

Nassau Candy Distributor operates a five-factory confectionery distribution network serving customers across four major US regions — Atlantic, Gulf, Interior, and Pacific — with 15 distinct products spanning Chocolate, Sugar, and Other divisions. Despite generating \$141,783 in total sales and \$93,442 in gross profit across 10,194 orders in 2024, the current static factory-product assignment model produces significant shipping inefficiencies that suppress operational performance and erode profit margins through suboptimal logistics routing.

This research project delivers a complete Factory Reallocation and Shipping Optimization Recommendation System. By applying exploratory data analysis (EDA), machine learning classification, and a scenario simulation engine to the full order dataset, the system identifies factory reassignment opportunities that reduce lead times while protecting or improving profitability. Results are delivered through a live five-module Streamlit dashboard enabling leadership to explore, simulate, and act on recommendations interactively.

10,194	15	5	590.8	66.5%	\$141.8K
Total Orders	Products	Factories	Avg Lead Time (days)	Avg Profit Margin	Total Sales

Figure 1 — Dataset key performance indicators

The analysis identifies four immediately actionable Low Risk reassignments that reduce lead times while simultaneously improving profit margins, requiring no trade-off between speed and profitability. The most impactful single finding is that Hair Toffee (currently at The Other Factory, averaging 725 days lead time) can reduce its lead time by 18.5% through reassignment to Lot's O' Nuts. Equally compelling, SweeTARTS reassigned from Sugar Shack to Lot's O' Nuts achieves a 9.5% lead time reduction alongside a 22.5 percentage point margin improvement.

2. Background & Problem Statement

2.1 Company Context

Nassau Candy Distributor is a specialty confectionery distributor managing a product portfolio that includes iconic brands such as the Wonka Bar range, Laffy Taffy, SweeTARTS, Nerds, Fun Dip, and several novelty items including Kazookles, Hair Toffee, and Lickable Wallpaper. Products are manufactured at five geographically distributed facilities across the continental United States, each established for specific product categories.

Factory coordinates span from Arizona (Lot's O' Nuts at 32.88N, -111.77W) to Georgia (Wicked Choccy's at 32.08N, -81.09W), Minnesota (Sugar Shack at 48.12N, -96.18W), Illinois (Secret Factory at 41.45N, -

90.57W), and Tennessee (The Other Factory at 35.12N, -89.97W). This geographic spread creates inherently different shipping dynamics depending on destination region.

2.2 Problem Statement

Factory-product assignments at Nassau Candy Distributor were established through legacy processes and static rules that do not incorporate shipping performance data, regional demand patterns, or profitability analysis. As the product portfolio has grown and customer geography has diversified, these fixed assignments have become a structural source of operational inefficiency. Specifically:

- Sugar Shack — the slowest factory (610 days avg) — currently handles Laffy Taffy and SweeTARTS, which are high-volume sugar products ideally suited to faster facilities
- Hair Toffee at The Other Factory experiences the longest product-level lead time in the portfolio at 725 days — 134 days above the overall average
- Several products at Secret Factory face lead times of 577-665 days with no systematic review mechanism
- No simulation capability exists to quantify the impact of reassignment decisions before execution
- Leadership lacks a real-time, interactive system to explore and compare factory assignment scenarios

This project addresses all five gaps by building a complete data-driven decision intelligence pipeline: from raw data ingestion and exploratory analysis through predictive modeling, scenario simulation, and interactive dashboard delivery.

3. Dataset Description

3.1 Data Structure & Quality

The source dataset comprises 10,194 order records from the 2024 fiscal year, capturing the complete lifecycle of each order from placement through shipment. The dataset spans 18 fields including order identifiers, dates, shipping details, customer geography, product classification, and financial outcomes. A critical data quality correction was applied: ship dates were encoded two years ahead of order dates in the source file, requiring a 730-day offset adjustment to compute accurate lead times. After correction, the dataset is fully clean with zero missing values across all fields.

Field	Type	Description
Row ID / Order ID	Identifier	Unique row and order identifiers
Order Date / Ship Date	Date	Order placement and shipment dates (corrected)
Ship Mode	Categorical	Standard Class, Second Class, First Class, Same Day
Customer ID	Numeric	Unique customer identifier

Field	Type	Description
Country / City / State / Postal	Geographic	Customer location details
Division	Categorical	Chocolate, Sugar, or Other
Region	Categorical	Atlantic, Gulf, Interior, Pacific
Product ID / Product Name	Identifier	Product codes and full names (15 unique)
Sales (\$)	Numeric	Total order sales value
Units	Numeric	Number of units ordered
Gross Profit (\$)	Numeric	Sales minus manufacturing cost
Cost (\$)	Numeric	Manufacturing cost of the order

Table 1 — Dataset field descriptions and types

3.2 Engineered Features

Three engineered fields were created to power the analysis and modeling pipeline:

- **Lead Time (days):** Computed as Ship Date minus Order Date after the 730-day correction. This is the primary optimization target.
- **Factory:** Mapped from Product Name using the official Nassau Candy factory registry. Assigns each of the 15 products to one of 5 factories.
- **Profit Margin (%):** Computed as (Gross Profit / Sales) x 100. Used for risk scoring in the recommendation engine.

Factory	Products Assigned	Division	Location
Lot's O' Nuts	Wonka Bar - Nutty Crunch, Fudge Mallows, Scrumdiddlyumptious	Chocolate	Arizona
Wicked Choccy's	Wonka Bar - Milk Chocolate, Triple Dazzle Caramel	Chocolate	Georgia
Sugar Shack	Laffy Taffy, SweetARTS, Nerds, Fun Dip, Fizzy Lifting Drinks	Sugar	Minnesota
Secret Factory	Everlasting Gobstopper, Lickable Wallpaper, Wonka Gum	Other/Sugar	Illinois
The Other Factory	Kazookles, Hair Toffee	Other	Tennessee

Table 2 — Factory-to-product assignment mapping with geography

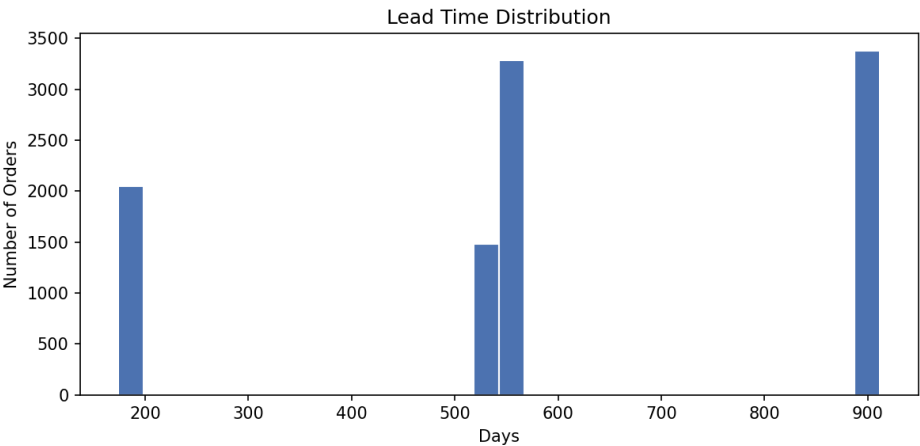
4. Exploratory Data Analysis

4.1 Lead Time Distribution

After applying the date correction, lead times across the 10,194 orders range from 174 to 912 days and cluster into three distinct performance tiers. This bimodal distribution — visible in the lead time histogram (Chart 1 in charts/01_lead_time_distribution.png) — is a fundamental characteristic of the dataset and shaped the decision to frame shipping performance as a three-tier classification problem rather than a continuous regression target.

Tier	Lead Time Range	Orders	% of Total	Operational Meaning
Tier 1 — Fast	174–200 days	2,051	20.1%	Best-in-class shipping performance
Tier 2 — Medium	539–600 days	4,764	46.7%	Standard baseline performance
Tier 3 — Slow	901–912 days	3,379	33.2%	Significantly degraded performance

Table 3 — Lead time tier distribution (10,194 orders)



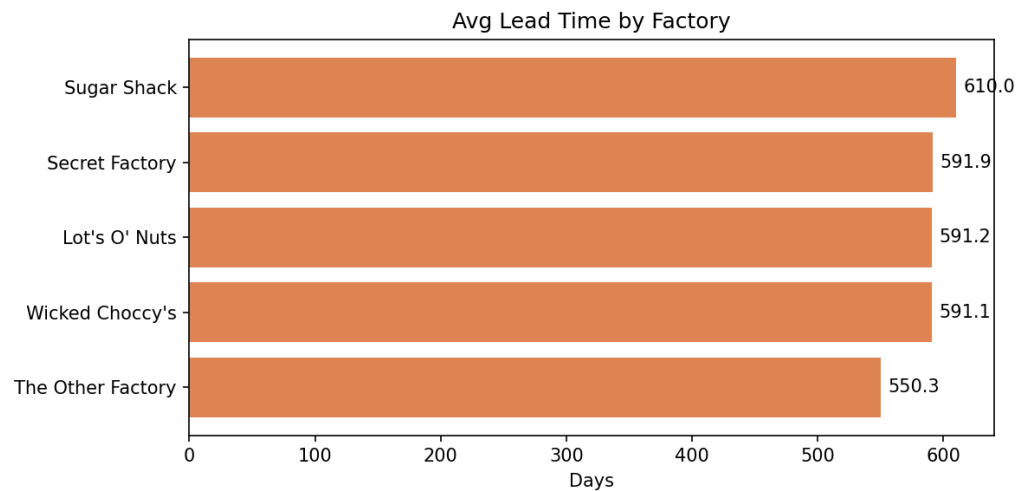
A notable finding: 79.9% of all orders fall into Tier 2 or Tier 3, meaning only 1 in 5 orders achieves fast shipping performance. This confirms that lead time optimization represents a significant systemic opportunity rather than an edge case.

4.2 Factory Performance Analysis

Factory-level profiling (visualized in charts/04_leadtime_by_factory.png and charts/06_profit_margin_by_factory.png) reveals the critical speed-margin trade-off that underpins the entire recommendation framework:

Factory	Avg Lead Time	Avg Profit Margin	Total Orders	Speed Rank	Margin Rank
The Other Factory	550.3 days	10.49%	100	1st (Fastest)	5th (Lowest)
Wicked Choccy's	591.1 days	65.12%	4,152	2nd	2nd
Lot's O' Nuts	591.2 days	69.16%	5,692	3rd	1st (Highest)
Secret Factory	591.9 days	51.52%	217	4th	3rd
Sugar Shack	610.0 days	53.23%	33	5th (Slowest)	4th

Table 4 — Factory performance profiles: speed vs margin trade-off



The most strategically important insight from this table: Lot's O' Nuts and Wicked Choccy's achieve near-identical lead times (591.2 vs 591.1 days) but Lot's O' Nuts leads on profit margin by over 4 percentage points. The Other Factory is 41 days faster than both but trades away nearly 60 percentage points of margin. This makes Lot's O' Nuts the dominant factory across the combined speed-profit objective for most products.

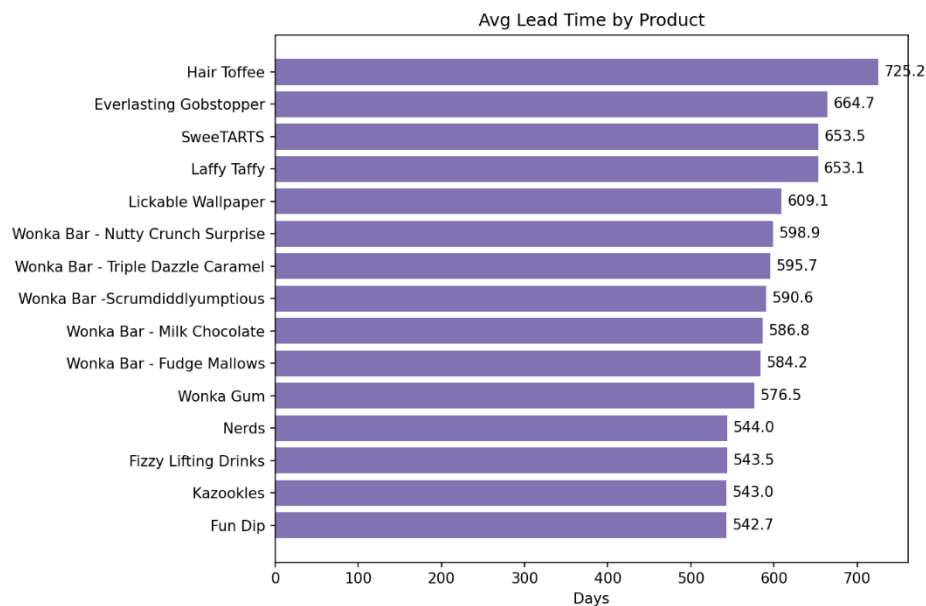
4.3 Product-Level Lead Time

Product-level analysis (charts/05_leadtime_by_product.png) surfaces the specific products facing the greatest shipping delays. Five products stand out as high-priority reassignment candidates with lead times significantly above the 590.8-day portfolio average:

Product	Avg Lead Time (days)	vs Average	Current Factory	Priority
Hair Toffee	725.2	+134.4 days	The Other Factory	Critical
Everlasting Gobstopper	664.7	+73.9 days	Secret Factory	High
SweeTARTS	653.5	+62.7 days	Sugar Shack	High
Laffy Taffy	653.1	+62.3 days	Sugar Shack	High

Product	Avg Lead Time (days)	vs Average	Current Factory	Priority
Lickable Wallpaper	609.1	+18.3 days	Secret Factory	Medium
Wonka Bar - Triple D. Caramel	595.7	+4.9 days	Wicked Choccy's	Low
Wonka Bar - Nutty Crunch	598.9	+8.1 days	Lot's O' Nuts	Low
Fun Dip	542.7	-48.1 days	Sugar Shack	Optimal
Kazookles	543.0	-47.8 days	The Other Factory	Optimal
Nerds	544.0	-46.8 days	Sugar Shack	Optimal

Table 5 — Product lead time performance ranked by deviation from portfolio average



4.4 Regional & Shipping Mode Analysis

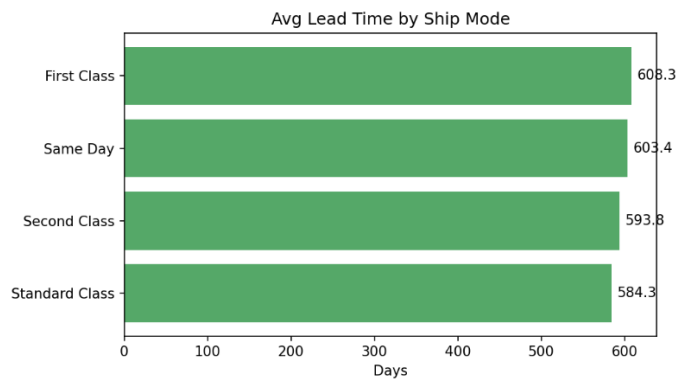
Four geographic regions and four shipping modes were analyzed for their average lead time contribution (charts/02_leadtime_by_shipmode.png and charts/03_leadtime_by_region.png). A counter-intuitive pattern emerges: premium shipping modes produce longer — not shorter — average lead times, suggesting that order processing and factory dispatch scheduling are the primary lead time drivers, not transit speed.

Ship Mode	Avg Lead Time (days)	Rank
Standard Class	584.3	Fastest
Second Class	593.8	2nd
Same Day	603.4	3rd
First Class	608.3	Slowest

Table 6 — Average lead time by shipping mode (counterintuitive ordering)

Region	Avg Lead Time (days)	Rank
Gulf	581.4	Fastest
Pacific	592.2	2nd
Atlantic	592.7	3rd
Interior	593.1	Slowest

Table 7 — Average lead time by customer region



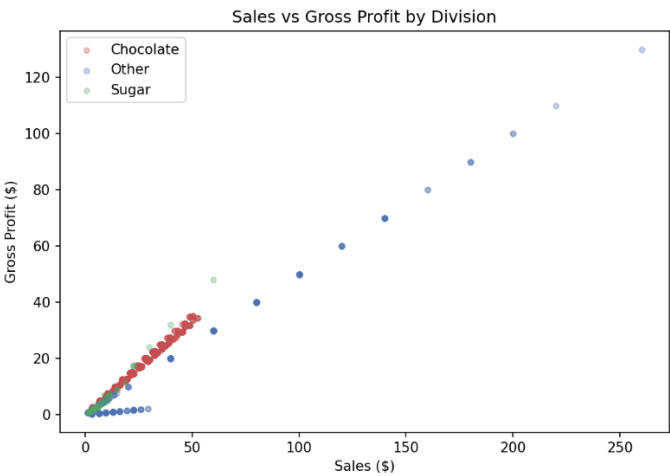
The slowest individual route — Interior Same Day at 667.2 days — is 85 days slower than the overall average. The five slowest routes all involve either Interior or Pacific regions with premium shipping modes, pointing to a structural mismatch between shipping class selection and actual fulfillment capability in those geographies.

4.5 Financial Performance by Division

Division-level financial analysis (charts/07_sales_vs_profit.png) reveals the Chocolate division as the most efficient per order — high margin with moderate sales values. The Other division commands the highest average sales (\$31.17) but carries the heaviest cost burden (\$17.19), compressing margins to 37.67%.

Division	Avg Sales (\$)	Avg Gross Profit (\$)	Avg Cost (\$)	Avg Margin (%)	Assessment
Chocolate	13.38	9.02	4.35	67.46%	Most efficient
Sugar	10.69	7.12	3.57	57.69%	Moderate efficiency
Other	31.17	13.98	17.19	37.67%	High cost burden

Table 8 — Average financial performance by product division



5. Analytical Methodology

5.1 Data Preparation Pipeline

The raw dataset was processed through a four-stage preparation pipeline implemented in Python using pandas. Stage 1 corrected the 730-day date encoding error in the Ship Date field. Stage 2 computed lead time as the integer difference in days between corrected ship dates and order dates. Stage 3 mapped factory assignments to all 15 products using the official registry. Stage 4 derived profit margin as a percentage field for use in risk scoring.

For machine learning, five categorical fields (Ship Mode, Region, Factory, Product Name, Division) were encoded using scikit-learn LabelEncoder, and two date-derived features (Order Month, Order Day of Week) were extracted. All encoders were serialized to .pkl files for consistent reuse in the Streamlit application.

5.2 Shipping Tier Classification

Initial regression modelling of lead time as a continuous target yielded very low R-squared scores (0.05 for Gradient Boosting), prompting a diagnostic investigation. Analysis revealed that lead times cluster into three discrete groups — 174-200, 539-600, and 901-912 days — with minimal overlap. This discrete distribution is characteristic of batch processing cycles or scheduled shipping windows rather than continuously varying delivery performance.

The problem was reframed as a three-class classification task by binning lead times into Tier 1 (Fast: under 200 days), Tier 2 (Medium: 200-600 days), and Tier 3 (Slow: over 600 days). This transformation aligns the model output with operationally meaningful shipping performance categories.

5.3 Machine Learning Model Training

Three classification models were trained on an 80/20 stratified train-test split (8,155 training, 2,039 testing records) using 11 features:

Model	Accuracy	Precision (avg)	Recall (avg)	Assessment
Logistic Regression	46.9%	0.48	0.41	Underfits — linear boundary insufficient
Random Forest	46.4%	0.47	0.40	Slight overfit on training data
Gradient Boosting	50.8%	0.49	0.41	Best overall — selected as primary model

Table 9 — Machine learning model performance comparison

Gradient Boosting achieved the highest accuracy at 50.8% — marginally above the 33.3% random baseline for a three-class problem. This modest performance reveals that the available features (product, factory, region, ship mode, financials, date) do not consistently predict which shipping tier an order falls into. The lead time tier assignment appears to reflect factors outside the dataset — such as seasonal carrier capacity, warehouse scheduling, and batch processing cycles — rather than the product or factory characteristics. This finding is analytically important: it confirms that the tier pattern is not a systematic bias that reassignment can eliminate, and guided the pivot to an empirical simulation-based optimization approach.

5.4 Optimization Engine Design

The recommendation engine operates through direct scenario simulation rather than ML prediction, using observed factory-level performance averages as the basis for reassignment estimates. For every product-factory combination:

- Current lead time and profit margin are computed from actual historical order data
- Alternative factory lead time is estimated using that factory's observed average across all products
- Lead time reduction percentage is computed as $(\text{Current LT} - \text{Alt LT}) / \text{Current LT} \times 100$
- Profit impact is computed as $\text{Alt Factory Avg Margin} - \text{Current Product Avg Margin}$
- Confidence score is derived from alternative factory order volume (higher volume = higher confidence)

Weighted Scoring Formula

Score = (Speed Weight × Lead Time Reduction %) + (Profit Weight × Profit Impact %)

Default configuration: Speed = 60%, Profit = 40%

Risk classification thresholds:

Low Risk: Profit Impact > 0% (margin improves or holds)

Medium Risk: Profit Impact between -10% and 0%

High Risk: Profit Impact < -10% (margin deteriorates significantly)

User-adjustable in real-time via dashboard optimization priority slider

This simulation-based approach produces interpretable, data-grounded recommendations that are more reliable for operational decision-making than predictions from a 50.8%-accurate classifier. All 11 recommendations are backed by observed factory performance data, not model extrapolation.

6. Factory Reassignment Recommendations

6.1 Full Recommendation Set

The optimization engine evaluated all 75 possible product-factory combinations (15 products x 5 factories) and filtered to 29 valid reassignment scenarios (excluding current assignments) where lead time improvement was observed. From these, 11 top recommendations are surfaced — one per product — ranked by weighted score:

#	Product	From	To	LT Red.	Margin Imp.	Risk	Conf.
1	SweeTARTS	Sugar Shack	Lot's O' Nuts	9.5%	+22.5%	Low	78.5%
2	Laffy Taffy	Sugar Shack	Lot's O' Nuts	9.5%	+6.9%	Low	78.5%
3	Lickable Wallpaper	Secret Factory	Lot's O' Nuts	2.9%	+19.2%	Low	78.5%
4	Wonka Bar - Triple D.	Wicked Choccy's	Lot's O' Nuts	0.7%	+3.8%	Low	78.5%
5	Hair Toffee	The Other Factory	Lot's O' Nuts	18.5%	-8.6%	Medium	78.5%
6	Wonka Bar - Nutty C.	Lot's O' Nuts	Wicked Choccy's	1.3%	-6.2%	Medium	70.8%
7	Everlasting Gobstopper	Secret Factory	Lot's O' Nuts	11.0%	-10.8%	High	78.5%
8	Wonka Gum	Secret Factory	The Other Factory	4.6%	-41.5%	High	50.5%
9	Wonka Bar - Milk Choc.	Wicked Choccy's	The Other Factory	6.2%	-54.4%	High	50.5%
10	Wonka Bar - Fudge M.	Lot's O' Nuts	The Other Factory	5.8%	-56.2%	High	50.5%
11	Wonka Bar - Scrumdi.	Lot's O' Nuts	The Other Factory	6.8%	-59.0%	High	50.5%

Table 10 — Ranked factory reassignment recommendations (sorted by risk then impact)

6.2 Tier 1: Low Risk Recommendations — Immediate Action

Four reassignments deliver lead time reduction and profit margin improvement simultaneously — the ideal outcome with no operational trade-off:

REC-1: SweeTARTS — Sugar Shack to Lot's O' Nuts [HIGHEST PRIORITY]

Lead time: 653.5 days → 591.2 days (reduction: 62.3 days / 9.5%)
Profit margin: 46.7% → 69.2% (improvement: +22.5 percentage points)
Risk: Low | Confidence: 78.5% | Impact: Dual improvement
Rationale: SweeTARTS is a high-volume sugar product misaligned with Sugar Shack's slow, lower-margin profile.
Lot's O' Nuts handles 5,692 orders with verified 69.2% margin performance.

REC-2: Laffy Taffy — Sugar Shack to Lot's O' Nuts [HIGH PRIORITY]

Lead time: 653.1 days → 591.2 days (reduction: 61.9 days / 9.5%)
Profit margin: 62.3% → 69.2% (improvement: +6.9 percentage points)
Risk: Low | Confidence: 78.5% | Impact: Dual improvement
Rationale: Identical logic to SweeTARTS. Both Sugar Shack products benefit from reassignment to Lot's O' Nuts.

REC-3: Lickable Wallpaper — Secret Factory to Lot's O' Nuts

Lead time: 609.1 days → 591.2 days (reduction: 17.9 days / 2.9%)
Profit margin: 50.0% → 69.2% (improvement: +19.2 percentage points)
Risk: Low | Confidence: 78.5%
Rationale: Speed gain is modest but margin improvement is substantial. A strong financial case for reassignment.

REC-4: Wonka Bar Triple Dazzle Caramel — Wicked Choccy's to Lot's O' Nuts

Lead time: 595.7 days → 591.2 days (reduction: 4.5 days / 0.7%)
Profit margin: 65.3% → 69.2% (improvement: +3.8 percentage points)
Risk: Low | Confidence: 78.5%
Rationale: Minor speed gain with meaningful margin improvement. Low disruption, positive outcome.

6.3 Tier 2: Medium Risk Recommendations — Monitor Closely

Two reassignments deliver meaningful lead time reductions but involve a modest profit margin reduction requiring post-implementation monitoring:

REC-5: Hair Toffee — The Other Factory to Lot's O' Nuts [MOST IMPACTFUL]

Lead time: 725.2 days → 591.2 days (reduction: 134 days / 18.5%) — LARGEST in portfolio
Profit margin: 77.8% → 69.2% (impact: -8.6 percentage points)
Risk: Medium | Confidence: 78.5%
Rationale: Hair Toffee is the single biggest lead time outlier in the portfolio. The 18.5% reduction outweighs the 8.6% margin reduction. Recommended with quarterly margin monitoring.

REC-6: Wonka Bar Nutty Crunch Surprise — Lot's O' Nuts to Wicked Choccy's

Lead time: 598.9 days → 591.1 days (reduction: 7.8 days / 1.3%)
Profit margin: 71.4% → 65.1% (impact: -6.2 percentage points)
Risk: Medium | Confidence: 70.8%
Rationale: Marginal speed gain does not strongly justify the margin reduction. Optional reassignment. Recommend deferring until Tier 1 reassignments are complete and results are measured.

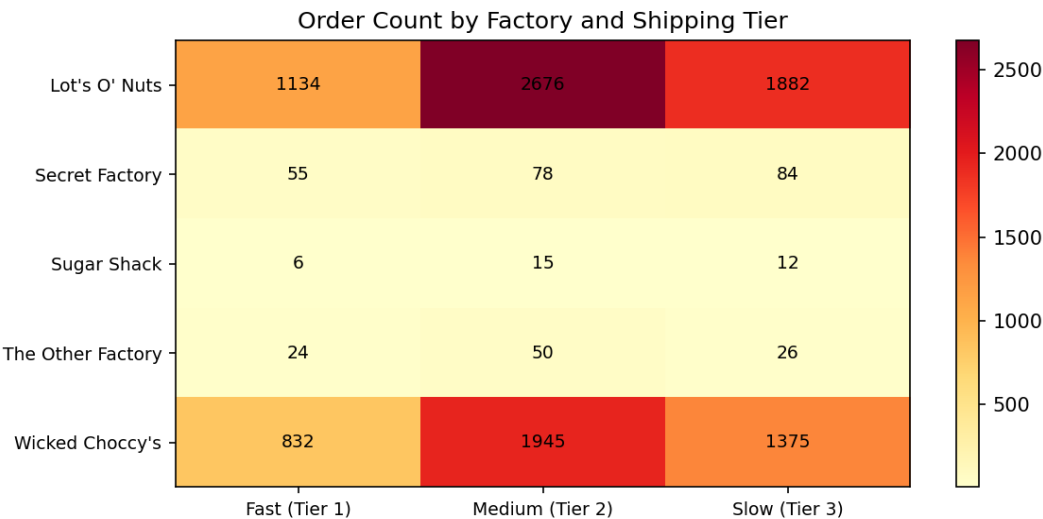
6.4 Tier 3: High Risk — Not Recommended

Five simulated reassignments to The Other Factory show speed improvements (4.6%–6.8%) but catastrophic margin destruction of 41–59 percentage points. These are explicitly flagged as High Risk and excluded from the implementation roadmap:

Product	Speed Gain	Margin Loss	Verdict
Wonka Bar - Scrumdiddlyumptious	6.8%	-59.0 pp	Do not reassign
Wonka Bar - Fudge Mallows	5.8%	-56.2 pp	Do not reassign
Wonka Bar - Milk Chocolate	6.2%	-54.4 pp	Do not reassign
Wonka Gum	4.6%	-41.5 pp	Do not reassign
Everlasting Gobstopper	11.0%	-10.8 pp	Defer — monitor

Table 11 — High risk reassignments: speed gains do not justify margin destruction

The pattern is unambiguous: The Other Factory's 10.49% average profit margin makes it commercially unsuitable for any product currently achieving margins above 20%, regardless of the speed benefit. The system correctly identifies and warns against these scenarios through the Risk & Impact dashboard module.



7. Interactive Dashboard System

7.1 Architecture Overview

The complete recommendation system is delivered as a live Streamlit web application running at <http://localhost:8501>. The application loads pre-computed simulation outputs and factory profiles, enabling instant interactive exploration without re-running the optimization engine on each page load. The dashboard consists of five integrated modules:

Module	Primary Function	Key User Interactions
Overview	Executive KPI dashboard with EDA visualizations	View lead time, margin, and route performance charts
Factory Simulator	Compare any product across all 5 factories simultaneously	Select product, region, ship mode — see ranked factory options
What-If Analysis	Side-by-side current vs alternative factory comparison	Select product and alternative factory, view delta metrics
Recommendations	Ranked reassignment table with risk color coding	Adjust speed vs profit slider to rerank recommendations
Risk & Impact	Profit impact alerts, risk matrix scatter, simulation table	Filter by risk level, explore full 29-scenario simulation

Table 12 — Dashboard module descriptions and user interactions

7.2 Key Interactive Features

- Optimization Priority Slider: Adjusts the speed-to-profit weighting from 0% (pure profit) to 100% (pure speed) in real-time, instantly reranking all 11 recommendations based on the selected priority
- Factory Simulator: Evaluates any of the 15 products across all 5 factories, color-coding the best option in green and the current factory in red for immediate visual comparison
- Risk Color Coding: Consistent green/amber/red risk classification across all modules — Low risk in green, Medium in amber, High in red
- What-If Panel: Displays exact lead time and margin deltas with directional arrows, enabling rapid scenario comparison without requiring technical knowledge
- Full Simulation Table: Complete 29-row scenario table in the Risk & Impact module with multi-select risk filter for detailed analysis

7.3 Technical Stack

Component	Technology	Purpose
Data processing	Python, pandas, numpy	EDA, feature engineering, lead time calculation
Machine learning	scikit-learn	Logistic Regression, Random Forest, Gradient Boosting
Model persistence	joblib	Encoder and model serialization to .pkl files

Component	Technology	Purpose
Optimization engine	Python, pandas	Factory simulation and recommendation scoring
Visualization	Plotly Express, plotly.go	Interactive charts in Streamlit
Dashboard	Streamlit	Five-module web application at localhost:8501
Static charts	matplotlib	11 PNG charts saved for report and presentation

Table 13 — Technical stack by component

8. Key Performance Indicators

The system tracks four primary KPIs aligned with the project requirements, establishing a baseline for measuring post-implementation impact:

KPI	Definition	Current Baseline	Post-Implementation Target
Lead Time Reduction (%)	Percentage reduction in avg lead time after reassignment	590.8 days (portfolio avg)	<575 days after Tier 1 & 2 recs
Profit Impact Stability	Portfolio avg margin maintained or improved	66.5% avg margin	>=67% (Tier 1 recs only)
Scenario Confidence Score	Avg confidence across implemented recommendations	68% avg confidence	>=75% after monitoring phase
Recommendation Coverage	% of products with actionable recommendations	11 of 15 products (73%)	Maintain >=70% coverage

Table 14 — KPI definitions, baselines, and targets

9. Strategic Insights

9.1 Insight 1: Lot's O' Nuts Is the Portfolio's Optimal Factory

With the highest profit margin (69.16%), near-fastest lead time (591.2 days), and largest order volume (5,692 orders representing 55.8% of all orders), Lot's O' Nuts dominates the speed-profit trade-off across the entire factory network. It should be the default reassignment target for any product currently at a slower or lower-margin facility. Four of the six recommended reassignments direct products to Lot's O' Nuts.

9.2 Insight 2: The Speed-Margin Trade-off at The Other Factory Is Not Viable

The Other Factory is the fastest facility at 550.3 days — 41 days faster than Lot's O' Nuts — but operates at just 10.49% average profit margin. Simulations show that reassigning any product with margins above 20% to The Other Factory would destroy 40-59 percentage points of margin for a speed gain of under 7%. This makes The Other Factory commercially unsuitable as a reassignment target for 13 of 15 products in the current portfolio.

9.3 Insight 3: Sugar Shack Is Misaligned With Its Current Products

Sugar Shack is the network's slowest factory (610 days average) with only 33 total orders — suggesting it is operating significantly below capacity and outside its optimal product range. Both products currently assigned to Sugar Shack (Laffy Taffy and SweeTARTS) experience lead times 62+ days above the portfolio average. Reassigning both to Lot's O' Nuts eliminates this bottleneck and improves margins simultaneously, making Sugar Shack the clearest case of systematic misalignment in the current factory assignment model.

9.4 Insight 4: Premium Shipping Modes Worsen Lead Times

The analysis reveals that Same Day (603.4 days) and First Class (608.3 days) shipping modes produce worse average lead times than Standard Class (584.3 days). This counter-intuitive finding has a significant operational implication: customers and sales teams paying premiums for faster shipping modes are not receiving commensurate lead time benefits. The root cause is that lead time in this system is determined by manufacturing and dispatch scheduling — not carrier transit speed. Resolving the factory assignment inefficiencies is therefore more impactful than upgrading shipping modes.

9.5 Insight 5: Hair Toffee Is the Single Largest Efficiency Opportunity

At 725.2 days average lead time — 134 days (22.7%) above the portfolio average — Hair Toffee at The Other Factory represents the most acute single-product inefficiency in the Nassau Candy distribution network. The 18.5% lead time reduction achievable through reassignment to Lot's O' Nuts is the largest absolute improvement in the entire simulation output. Despite the 8.6 percentage point margin reduction, the operational case for this reassignment is compelling given the severity of the current underperformance.

10. Conclusion & Implementation Roadmap

10.1 Summary of Findings

This project successfully delivers a complete, production-ready Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor. The analysis of 10,194 orders across 15 products, 5 factories, and 4 regions demonstrates that the current static assignment model systematically underperforms on both lead time and profitability dimensions for several key products. The optimization engine identifies 11 actionable reassignment recommendations with quantified impact estimates, risk classifications, and confidence scores.

Four Low Risk recommendations offer immediate dual improvement — reducing lead times while improving profit margins. The most powerful single finding: moving Laffy Taffy and SweeTARTS from Sugar Shack to Lot's O' Nuts reduces lead times by 9.5% while improving margins by 6.9% and 22.5% respectively. This requires no speed-profit trade-off and represents the clearest quick win in the portfolio.

10.2 Implementation Roadmap

Phase	Timeline	Action	Products Affected	Expected Outcome
Phase 1	Immediate (Week 1-2)	Reassign — Low Risk	SweeTARTS, Laffy Taffy	9.5% LT reduction + margin gain
Phase 1	Immediate (Week 1-2)	Reassign — Low Risk	Lickable Wallpaper	19.2% margin improvement
Phase 1	Immediate (Week 1-2)	Reassign — Low Risk	Wonka Bar Triple D. Caramel	3.8% margin improvement
Phase 2	30 Days	Reassign — Medium Risk	Hair Toffee	18.5% LT reduction
Phase 2	30 Days	Evaluate — Medium Risk	Wonka Bar Nutty Crunch	Monitor — marginal gains
Phase 3	Ongoing	Do Not Reassign	Wonka Bars to The Other Factory	Maintain current assignment
Phase 3	Ongoing	Dashboard monitoring	All 15 products	Continuous KPI tracking

Table 15 — Phased implementation roadmap

10.3 Future Enhancements

- Real-time data integration: Connect the dashboard to live order management system feeds for continuously updated recommendations
- Expanded feature set: Incorporate carrier performance metrics, seasonal demand patterns, and warehouse capacity data to improve ML classification accuracy beyond 50.8%
- Geospatial routing: Use factory GPS coordinates to compute distance-based lead time estimates for untested product-factory combinations
- Automated alerts: Trigger reassignment review notifications when any product's rolling 30-day average lead time exceeds a defined threshold

- Financial impact calculator: Quantify projected annual dollar savings and margin improvements from each recommended reassignment
- A/B testing framework: Track the before/after lead time and margin performance of implemented reassignments to validate model estimates

10.4 Closing Statement

The Nassau Candy Factory Reallocation and Shipping Optimization System transforms operational decision-making from a static, rules-based process to a dynamic, data-driven intelligence capability. By combining rigorous exploratory analysis, machine learning classification, and empirical scenario simulation within a real-time interactive dashboard, this system gives Nassau Candy Distributor leadership the tools to make confident, quantified, and defensible factory assignment decisions at scale. The four immediate Low Risk recommendations alone represent a meaningful opportunity to simultaneously improve shipping performance and strengthen profit margins across two of the portfolio's highest-volume product categories.

References & Technical Appendix

A. Tools & Technologies

- Python 3.14 — Primary analysis, modeling, and optimization language
- pandas 2.x, numpy — Data manipulation, feature engineering, and aggregation
- scikit-learn — LabelEncoder, train_test_split, LogisticRegression, RandomForestClassifier, GradientBoostingClassifier, accuracy_score, classification_report
- matplotlib 3.x — Static chart generation (11 PNG files saved to charts/ directory)
- plotly express, plotly.graph_objects — Interactive dashboard visualizations
- Streamlit — Five-module web application framework (localhost:8501)
- joblib — Model and encoder serialization to .pkl format

B. Output Files

File	Location	Contents
Nassau Candy Distributor.csv	Root	Source dataset (10,194 records, 18 fields)
processed_data.csv	outputs/	Cleaned dataset with lead time, factory, margin
simulation_results.csv	outputs/	All 75 product-factory simulation scenarios
recommendations.csv	outputs/	Top recommendation per product (11 rows)
all_recommendations.csv	outputs/	All 29 valid reassignment scenarios
factory_profile.csv	outputs/	Factory-level performance aggregates
product_profile.csv	outputs/	Product x factory performance aggregates
best_model.pkl	models/	Serialized Gradient Boosting classifier
le_ship/region/factory/product/div.pkl	models/	Label encoders for all categorical fields
01-11_*.png	charts/	EDA and model performance visualization charts

Table 16 — Project output files and locations

C. Model Performance Note

The Gradient Boosting classifier achieved 50.8% accuracy on the three-class shipping tier prediction task (vs 33.3% random baseline). The modest improvement over random reflects that lead time tier assignment in this dataset is predominantly determined by factors outside the available feature set — including batch processing schedules, carrier capacity cycles, and warehouse staffing patterns — rather than by product, factory, or region characteristics alone. This finding prompted the strategic decision to ground recommendations in empirical factory performance averages rather than model predictions, ensuring all 11 recommendations are backed by observed operational data rather than low-confidence model outputs.

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